Review of Sets and Functions

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ECON 320 - Introduction to Mathematical Economics

A. Ingredients of a Mathematical Model

A mathematical model is a representation of an economic system using variables, equations, and assumptions to describe relationships and predict behavior.

Ingredients:

- Variables and Equations
- Functional Relationships
- Assumptions
- Parameters
- Boundary Conditions
- Objectives
- Constraints
- Equilibrium Conditions

1. Equations and Variables
Equations represent economic relationships and show how variables interact.

Example: Demand equation

$$Q_d = 100 - 2P$$

where Q_d is quantity demanded and P is price.

2. Functional Relationships
Functions describe how one variable depends on others. Can be linear, nonlinear, or multi-variable.

Example: Cobb-Douglas production function

$$Y = AL^{\alpha}K^{\beta}$$

where Y is output, L is labor, K is capital, and A, α, β are constants.

3. Assumptions
Simplify reality for tractability, e.g., rational behavior, perfect competition, or ceteris paribus.

Example: Perfect competition implies no single buyer/seller can influence market price.

4. Parameters
Constants defining relationships, estimated from data.

Example: In $Q_d = 100 - 2P$, the slope -2 is a parameter showing sensitivity of demand to price.

5. Boundary Conditions
Initial or boundary values required for solving models, especially differential equations.

Example: Capital accumulation model with $K(0) = K_0$ as initial condition.

6. Objectives
Models often optimize objectives like profit, utility, or cost.

Example: Profit maximization:

$$\pi = P \cdot Q - C(Q)$$

where P is price, Q is quantity, $C(Q)$ is cost.

7. Constraints
Limitations like resources, budget, or technology.

Example: Budget constraint:

$$P_x \cdot x + P_y \cdot y = I$$

where P_x, P_y are prices and I is income.

8. Equilibrium Conditions
States where economic forces balance, e.g., supply equals demand.

Example: Market equilibrium:

$$Q_s = Q_d$$

Conclusion
Mathematical models integrate these ingredients to analyze and predict economic behavior systematically.

B. Real Number System

The real number system is the foundation of mathematical analysis in economics. It includes both rational numbers (fractions) and irrational numbers (non-repeating, non-terminating decimals).

1. Classification of Numbers

- **Natural Numbers ($\mathbb{N}$):** $1, 2, 3, \ldots$
- **Whole Numbers ($\mathbb{W}$):** $0, 1, 2, 3, \ldots$
- **Integers ($\mathbb{Z}$):** $\ldots, -2, -1, 0, 1, 2, \ldots$
- **Rational Numbers ($\mathbb{Q}$):** numbers expressible as $\frac{p}{q}$, $q \neq 0$; e.g., $\frac{2}{3}, 0.75$
- **Irrational Numbers:** cannot be expressed as fractions; e.g., $\pi, \sqrt{2}$
- **Real Numbers ($\mathbb{R}$):** $\mathbb{R} = \mathbb{Q} \cup$ (irrationals)

2. Properties of Real Numbers

- **Algebraic:** closure, commutative, associative, distributive laws
- **Completeness:** every non-empty set bounded above has a least upper bound

Example of Completeness: The set $\{x \in \mathbb{Q} \mid x^2 < 2\}$ has no rational supremum, but its supremum in $\mathbb{R}$ is $\sqrt{2}$.

3. Intervals of Real Numbers

- **Closed Interval:** $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$
- **Open Interval:** $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$
- **Half-Open:** $[a, b)$ or $(a, b]$

4. Absolute Value

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Properties:

- $|xy| = |x||y|$
- $|x + y| \leq |x| + |y|$ (Triangle inequality)

5. Representation on the Number Line

Real numbers are points on a continuous line. Distance between a and b is $|a - b|$.

6. Applications in Economics

Real numbers allow modeling continuous variables: prices, quantities, interest rates, output. They are essential for derivatives, integrals, and continuous functions.

Examples of Real Numbers:

- Integers: $0, 7, -3$
- Rational: $\frac{5}{2}, -0.4$
- Irrational: $\pi, \sqrt{3}$
- All belong to $\mathbb{R}$

Understanding the real number system is fundamental for mathematical modeling in economics and provides the basis for analysis of continuous variables and functions.

C. The Concept of Sets

A set is a collection of objects called elements. Sets are foundational for grouping and modeling in economics.

1. Elements of a Set
- Objects contained in a set are called elements.
- Notation: $x \in A$ means x is in set A; $y \notin A$ means y is not in A.

Example:

$$A = \{apple, banana, orange\}, \quad apple \in A, \quad mango \notin A$$

2. Types of Sets

- **Finite Set:** Limited number of elements, e.g., $A = \{1, 2, 3\}$
- **Infinite Set:** Elements are unbounded, e.g., $B = \{1, 2, 3, \dots\}$
- **Empty Set:** Contains no elements, $\emptyset$ or $\{\}$
- **Singleton Set:** Contains exactly one element, $S = \{0\}$
- **Subsets:** $A \subseteq B$ means all elements of A are in B
- **Universal Set:** U contains all elements under consideration

3. Set Operations

- **Union ($\cup$):** $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- **Intersection ($\cap$):** $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- **Difference ($\setminus$):** $A \setminus B = \{x \mid x \in A, x \notin B\}$
- **Complement:** $A^c = U \setminus A$
- **Cartesian Product:** $A \times B = \{(a, b) \mid a \in A, b \in B\}$

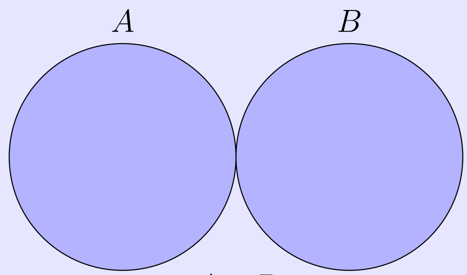
Example: Let $A = \{1, 2\}, B = \{2, 3\}$, $U = \{1, 2, 3, 4\}$

- $A \cup B = \{1, 2, 3\}$
- $A \cap B = \{2\}$
- $A \setminus B = \{1\}$
- $A^c = \{3, 4\}$
- $A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$

4. Venn Diagrams
A visual representation of sets and their relationships. Useful for unions, intersections, complements, and illustrating market overlaps, population segments, or preferences.
Let A and B be two sets. The following diagrams illustrate the fundamental set operations.

$A \cup B$ (Union)

Definition: The union of two sets A and B is the set of all elements that are in A, in B, or in both.

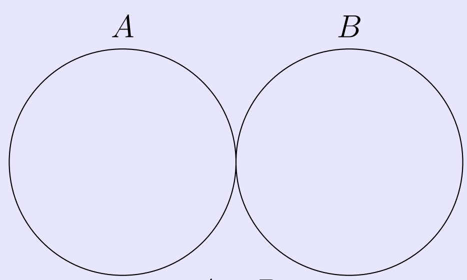


$A \cup B$

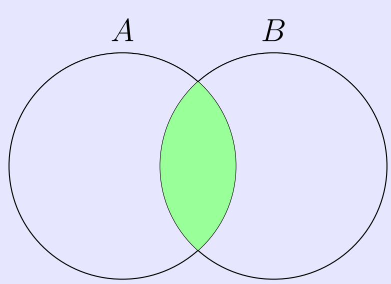
Explanation: All elements that belong to either A, B, or both are included. The entire area covered by both circles is shaded.

$A \cap B$ (Intersection)

Definition: The intersection of A and B is the set of elements that are in both A and B.



$A \cap B$

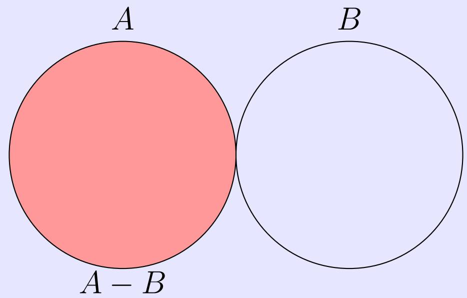


$A \cap B$

Explanation: Only elements that belong to **both** sets are included. The overlapping area is shaded.

$A - B$ (Difference)

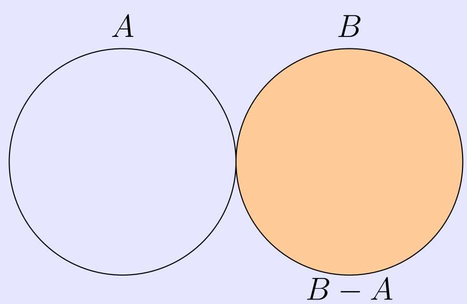
Definition: The difference $A - B$ is the set of elements that are in A but not in B.



Explanation: Elements unique to A are included; the overlapping region with B is excluded.

$B - A$ (Difference)

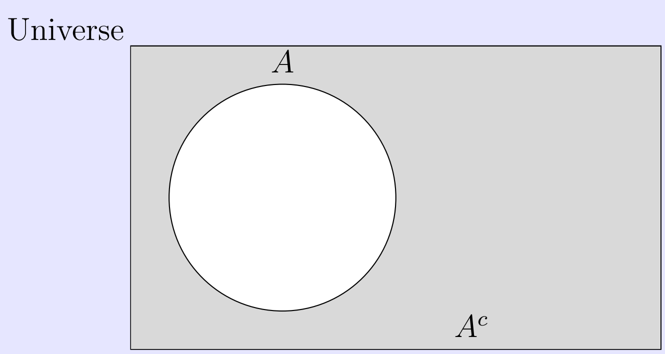
Definition: The difference $B - A$ is the set of elements that are in B but not in A.



Explanation: Elements unique to B are included; overlapping with A is excluded.

A^c (Complement)

Definition: The complement of A, denoted A^c, is the set of all elements not in A.



Explanation: The shaded area represents all elements **outside** set A, i.e., everything in the universe except A.

5. Properties of Set Operations

- **Commutative:** $A \cup B = B \cup A$, $A \cap B = B \cap A$
- **Associative:** $(A \cup B) \cup C = A \cup (B \cup C)$
- **Distributive:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **De Morgan's Laws:** $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$

6. Applications in Economics

- Consumer choice: set of all goods consumed
- Production: set of feasible production combinations
- Market analysis: set of firms, buyers, or products
- Probability and uncertainty: events as subsets of sample space

The concept of sets provides a fundamental framework for grouping elements, performing operations, and modeling economic phenomena systematically.

D. Relations and Functions

Relations and functions are fundamental concepts that describe how elements from one set are associated with elements of another. They are widely used in economics to model demand, supply, production, and preferences.

1. Relations

Definition: A relation R from set A to set B is a subset of the Cartesian product $A \times B$:

$$R \subseteq A \times B$$

Notation: If $(a, b) \in R$, we write $a\,R\,b$.

Example: Let $A = \{1, 2, 3\}$, $B = \{x, y\}$ Relation $R = \{(1, x), (2, y), (3, x)\}$

Types of Relations:

- Reflexive: aRa for all $a \in A$
- Symmetric: If aRb, then bRa
- Transitive: If aRb and bRc, then aRc
- Equivalence Relation: Reflexive, symmetric, and transitive
- Partial Order: Reflexive, antisymmetric, and transitive (e.g., $\leq$)

Economic Example: Preference relation $\succeq$ on consumption bundles: $(x_1, x_2) \in R$ if x_1 is at least as preferred as x_2.

2. Functions

A function f from set A (domain) to set B (codomain) is a relation where each element of A is related to exactly one element of B:

$$f : A \to B, \quad f(a) = b$$

Examples:

- $f(x) = 2x + 3, \quad x \in \mathbb{R}$
- Demand function: $Q_d(P) = 100 - 2P$
- Production function: $Y = F(K, L)$

Properties of Functions:

- Domain: input values
- Codomain: potential output values
- Range (Image): actual output values
- Monotonicity: increasing or decreasing
- Continuity: no abrupt jumps
- Differentiability: smooth changes

Examples:

- $f(x) = x^2, x \in \mathbb{R}$, Range: $[0, \infty)$
- Production function: $Y = K^{0.3}L^{0.7}, K, L \geq 0$

Classification of Functions:
2.1 According to Number of Variables:

- **Single-variable:** $y = f(x)$, e.g., $Q_d(P) = 100 - 2P$
- **Multivariable:** $z = f(x_1, x_2, \ldots, x_n)$, e.g., $Y = F(K, L)$

2.3 According to Continuity and Smoothness:

- **Continuous:** No jumps, e.g., $f(x) = x^2$
- **Discontinuous:** Has jumps, e.g., step functions
- **Differentiable:** Smooth, derivative exists, e.g., $f(x) = x^3$
- **Non-differentiable:** Sharp corners, e.g., $f(x) = |x|$ at $x = 0$

2.4 According to Nature of Function:

- **Linear:** $y = a + bx$, constant slope
- **Nonlinear:** $y = ax^2 + bx + c$, slope varies, e.g., Cobb-Douglas $Y = K^\alpha L^\beta$
- **Monotonic:** Increasing or decreasing
- **Homogeneous:** Scaling inputs scales output, e.g., $f(tK, tL) = t^r f(K, L)$
- **Explicit vs Implicit:** $y = f(x)$ (explicit), $F(x, y) = 0$ (implicit)

2.5 According to Output Mapping:

- **Real-Valued:** $f : \mathbb{R}^n \to \mathbb{R}$, e.g., $U(x_1, x_2) = x_1^{0.5}x_2^{0.5}$
- **Vector-Valued:** $f : \mathbb{R}^n \to \mathbb{R}^m$, e.g., market equilibrium mapping
- **Boolean:** Maps to 0 or 1, e.g., indicator functions

2.6 Special Functions in Economics: Step functions, piecewise functions, quadratic/polynomial functions, exponential and logarithmic functions.

Function Operations:

- Addition: $(f+g)(x) = f(x) + g(x)$
- Multiplication: $(f \cdot g)(x) = f(x) \cdot g(x)$
- Composition: $(f \circ g)(x) = f(g(x))$

Summary Table of Function Types

Type	**Definition**	**Economic Example**
Linear	Constant slope	Linear demand $Q = a - bP$
Nonlinear	Slope changes	Cobb-Douglas $Y = K^{\alpha}L^{\beta}$
Monotonic	Increasing/decreasing	Utility in goods
Injective	Unique mapping	Price-to-demand mapping
Surjective	Covers codomain	Production mapping
Bijective	One-to-one	Resource allocation mapping
Continuous	No jumps	Smooth cost function
Differentiable	Derivative exists	Production optimization
Homogeneous	Scale property	Returns to scale in production

Graphical Representation:

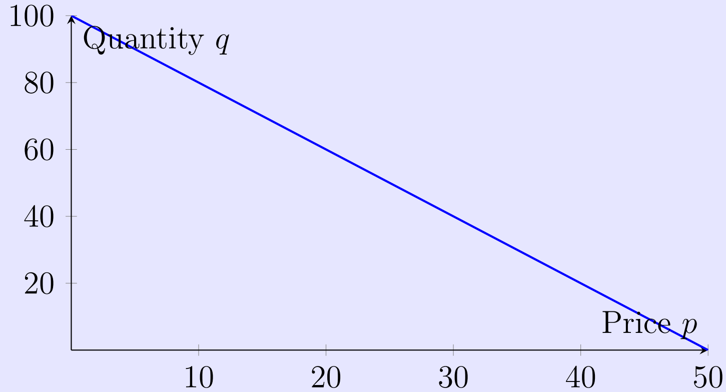
- Graph: set of points $(x, f(x))$
- Vertical line test: ensures each x has exactly one y

A function maps each element of one set (domain) to exactly one element of another set (codomain).

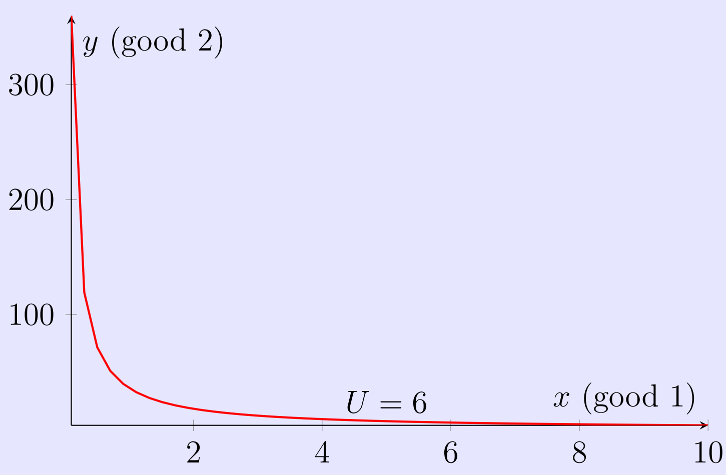
$$f : X \to Y$$

Example 1: Demand Function

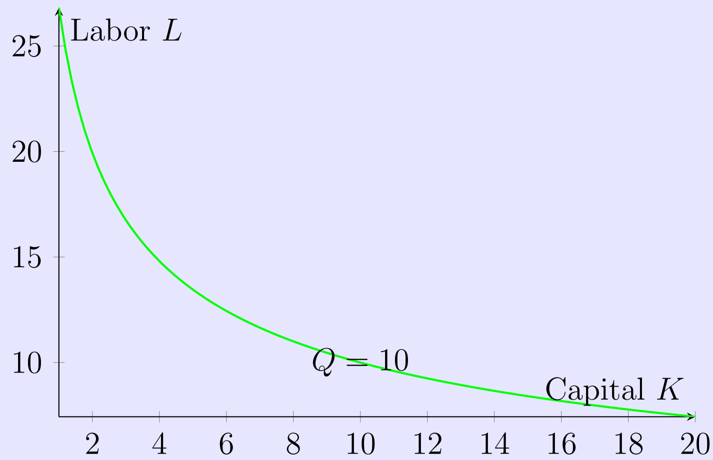
$$q(p) = 100 - 2p$$



Example 2: Utility Function: $U(x, y) = \sqrt{x}\sqrt{y}$ (Cobb–Douglas).



Example 3: Production Function: $Q(K, L) = K^{0.3}L^{0.7}$ (isoquant for $Q = 10$).



Applications in Economics:

- Consumer theory: Utility and demand functions
- Production theory: Production and cost functions
- Market analysis: Price-quantity mappings
- Growth models: Exponential/logarithmic growth
- Optimization: Monotonic and differentiable functions for maxima/minima

Functions and relations provide the mathematical framework for modeling economic relationships, linking variables systematically for analysis and prediction.

References

[1] Chiang, A. C. and Wainwright, K. (2005). *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw–Hill.

[2] Simon, C. P. and Blume, L. (2010). *Mathematics for Economists*. W. W. Norton.

[3] Sydsæter, K. and Hammond, P. (2022). *Essential Mathematics for Economic Analysis*, 6th ed., Pearson.